The Old-Age Tax and the Cap: Benchmarking the US Social-Security Levy Against Peer Pension Systems

TAXLANE Research — panel-reviewed (revised, round 2). An educational reform model. Figures are sourced to `docs/sources/source-version-ledger.md`; Trustees, SSA-actuary, and CBO figures are projections under stated assumptions. "The cap, not the rate" is a **normative** policy judgment (it rests on a progressivity-equals-fairness premise made explicit in §6), not a derived empirical finding. No tax advice is given.

Abstract

The US funds old-age and survivors insurance through a dedicated payroll tax, and the solvency debate usually pits "raise the rate" against "cut benefits." We argue for a third lever — **the wage cap** — and make a normative case that it is the fairest of the three. US public pension spending is **near the peer norm (7.3% of GDP vs the OECD's 8.1%)** [`SRC-OECD-PENSIONS-2025`]. The combined **OASDI rate (12.4%)** is below the large peer systems (Germany 18.6%, Japan 18.3%, Sweden 18.5%, Italy 33%) though above Canada's base CPP rate (11.9%) [`SRC-PENSION-CONTRIB-RATES`]. **Contrary to a common claim, the US is not unique in capping the taxed base** — Germany, Japan, Canada, and Sweden also cap — but the US combines a *below-peer rate* with a cap that exempts a *large and growing share* of earnings: **~17–18% of covered earnings are above the cap and untaxed** [`SRC-SSA-TRUSTEES-2026`]. The OASI trust fund is projected to deplete in **~2032** (combined OASDI ~2034; ~83% of benefits then payable, declining toward 65% by 2100); the 75-year actuarial deficit is **4.42% of taxable payroll** (2026 TR; −3.82% on the 2025 basis SSA uses to score provisions) [`SRC-SSA-TRUSTEES-2026`, `SRC-SSA-SOLVENCY`]. By SSA's actuarial scoring, **eliminating the cap closes ~67% of the 75-year shortfall (+2.55% of payroll) without a benefit credit, or ~48% with one**; taxing earnings over $250,000 closes ~65% [`SRC-SSA-SOLVENCY`]. Lifting the cap is progressive on an annual basis — though differential mortality complicates lifetime progressivity (§6) — whereas a rate increase is regressive and a benefit cut falls on retirees. The modernization case is to broaden the **base** while protecting benefits.

1. What the US funds today

Social Security is the federal budget's largest single lane: in FY2025 it cost **$1,580,673M** (OMB function 650 outlays, `SRC-OMB-HIST-3-2-FY2027`), funded by **$1,283,736M** of dedicated OASDI payroll receipts (`SRC-OMB-HIST-2-4-FY2027`) — a ~$297B current-year gap covered by trust-fund drawdown. The tax is **12.4% combined** (6.2% + 6.2%) on wages up to the **taxable maximum** ($176,100 in 2025, $184,500 in 2026) [`SRC-SSA-OACT`]. Program cost is **5.3% of GDP**, projected to rise toward a ~6.9% peak near mid-century as the population ages, then partially recede [`SRC-SSA-TRUSTEES-2026`].

2. Peer benchmark

Spending (OECD, public-only, 2021 basis)

Country	Public pension spend, % GDP
Italy 16.1 · France 13.4 · Germany 10.8 · Japan 9.2 Sweden 8.0 · **OECD avg 8.1** · **US 7.3** · UK 7.1 · Canada 5.9	

Source: `SRC-OECD-PENSIONS-2025`. US public pension spending is **at/below the OECD average** — not high.

Contribution rates (national agencies, 2025–26; pillar and cap labeled)

Country	Combined rate	Pillar	Base
Italy	33%	single	mostly uncapped
Germany	18.6%	single	capped (€96,600)
Sweden	18.5%	multi	capped
Japan	18.3%	multi-tier	capped
France	15.45% + 2.42% uncapped (+AGIRC-ARRCO)	multi	partly capped
United States	**12.4%**	single	**capped ($176,100)**
Canada (CPP)	11.9% (+CPP2)	single	capped
United Kingdom	(NI, not pension-earmarked)	—	—

Source: `SRC-PENSION-CONTRIB-RATES`. Two honest qualifications: the rates mix single-pillar (Germany) and multi-pillar (France, Sweden) systems and so are not strict like-for-like; and **the US is above Canada**, so "below peers" is not universal. The defensible reading is narrower and sharper: the US applies a **lower rate** than the large single- and multi-pillar systems **to a capped base**, where peers that also cap (Germany, Japan) pair the cap with **substantially higher rates**.

3. The cap: a low, eroding base — not a unique feature

Capping is common; the US distinction is the **combination** of a below-peer rate with a cap that exempts a **large and growing** share of earnings. **~17–18% of US covered earnings are above the cap and untaxed** [`SRC-SSA-TRUSTEES-2026`], and that share has **risen** as top earnings outpaced the median, so the base has eroded over recent decades. The US therefore raises less from old-age contributions than its 12.4% headline implies — not because capping is unusual, but because the US caps a *low* base at a *low* rate.

4. The solvency path

[`SRC-SSA-TRUSTEES-2026`]: OASI reserve depletion **~Q4 2032** (∼78% of benefits payable then); the theoretical **combined OASDI** fund **~2034** (83% payable at depletion, declining toward 65% by 2100 as the worker-to-beneficiary ratio falls). DI is solvent across the 75-year window. The **75-year actuarial deficit is 4.42% of taxable payroll** (2026 TR; ∼1.5% of GDP). This is a structural, demographic shortfall.

5. All factors → the recommended lever: the cap, not the rate

- **Solvency.** The 75-year deficit must be closed; the capped high-end base is the largest untapped source.
- **Structural.** The cap's erosion is itself a cause of the shortfall.

- **International.** Peers that cap pair it with higher rates; the US can raise more from its base without exceeding peer *rates*.

By SSA's **actuarial-balance scoring** (improvement in the 75-year balance, % of taxable payroll; share of the shortfall closed on the 2025 TR basis) [SRC-SSA-SOLVENCY] — these are 75-year actuarial figures, *not* 10-year dollar scores divided by the deficit:

Cap provision	Δ 75-yr balance	Share of shortfall
Eliminate the cap, no benefit credit	+2.55% of payroll	~67%
Eliminate the cap, with benefit credit	+1.85%	~48%
Tax earnings over $250,000, no credit	+2.50%	~65%
Raise cap to cover 90% of earnings, no credit	+1.07%	~28%

(Shares are on the 2025 TR −3.82%-of-payroll basis; on the 2026 TR 4.42% basis the same provisions close a somewhat smaller fraction.) None raises the 12.4% rate. **Behavioral response is largest exactly where the yield is largest:** a near-doubling of the top marginal payroll wedge on uncapped earnings would induce some shifting of compensation toward capital or deferral; SSA scores embed assumptions about this, but realized yields are uncertain — a caveat that belongs beside the 67% figure, not in a footnote.

6. Distribution and incidence

The normative premise, stated. Calling the cap "the fair lever" rests on the judgment that a more **progressive** financing change is fairer. That is a value premise, not a derived result; the analysis below is what it rests on.

Annual incidence. On an annual basis the cap is the most regressive feature of the tax: above the maximum the marginal OASDI rate is zero, so a $1,000,000 earner pays the same as one at $184,500. Lifting the cap is therefore **annually progressive**; a rate increase is regressive-to-flat (all wages, capital exempt); a benefit cut falls on retirees.

Lifetime incidence and differential mortality. The deeper public-finance point complicates this. Social Security's true incidence is **lifetime**, and **higher-income people live materially longer** — the life-expectancy gap between the top and bottom 1% of income is **14.6 years for men and 10.1 for women** [SRC-CHETTY-2016] — so high earners collect benefits for more years. Accounting for differential mortality **erodes, and in projection reverses, the lifetime progressivity** of the system [SRC-NAS-2015]. This cuts *toward* the recommendation: the **no-benefit-credit** cap options (which tax the high-end earnings without raising those earners' future benefits) are both more progressive *and* close more of the gap — they do not hand longer-lived high earners a larger benefit. The benefit formula itself (progressive bend points on career-average earnings) is progressive on contributions but is partly offset by the mortality gradient. A full decile/lifetime-incidence estimate is deferred pending microdata, but the direction is clear: cap reform **without** a benefit credit is the progressive choice.

7. The needs, the problem, and the modernization case

Why the US struggles here. A dedicated, contributory program is funded by a *low rate on a low, eroding base*, against an aging population — so the fund depletes on schedule while the headline rate looks "normal." The problem is **base erosion plus a below-peer rate**, not an unusually generous program.

The modernization case. Modernize the **base**: lift or eliminate the cap so the levy reaches high earnings as peer systems' higher rates effectively do, protect scheduled benefits, and stop treating a rate hike or a benefit cut as the only options. Keep the contributory promise by broadening *who* is taxed on *what*, not by taxing low earners more or paying retirees less.

8. Reform design: the cap as the lever

Social Security is the canonical **dedicated lane** — payroll in, benefits out, its own trust fund. The lane's operating rule is **"fix the base, not the rate"** for shortfalls: the `shortfall_rule` is satisfied by a cap adjustment with a published distributional record, not a rate increase or a silent benefit cut. `allocation_method = proposed_reform`. A trust-fund drawdown is financed by redeeming Treasuries — i.e., from public borrowing — so the "trust fund" is not outside the federal cash constraint (disclosed).

9. Discussion and limitations

- **Not a full solvency package.** Cap reform closes ~28–67% of the 75-year deficit depending on the option and benefit-credit choice; full solvency may also require benefit-formula or retirement-age changes (out of scope).
- **Benefit linkage.** Crediting the newly-taxed earnings raises future benefits and closes less; not crediting closes more and is more progressive but weakens the contributory link. The paper recommends the no-credit direction on progressivity grounds and flags the legitimacy tradeoff.
- **Behavioral response** is real and largest at the top; SSA scores embed assumptions, realized yields uncertain.
- **Comparability.** Peer contribution rates mix pillars and cap designs; they are labeled, not aggregated.
- **Projections.** Trustees/actuary figures are intermediate-assumption projections; depletion dates shifted 2033→2032 between the 2025 and 2026 reports.

10. Conclusion

The American old-age tax is not unusually generous, and capping is not unique to it; it is funded by a **below-peer rate on a low, eroding capped base**, so a sixth of earnings escapes, the fund depletes around 2032–34, and the 75-year deficit is 4.42% of payroll. On a normative progressivity premise, the fairest lever is the **cap, not the rate**: SSA scores eliminating it at ~67% of the shortfall, lifting it is annually progressive, and the no-credit version is the progressive choice once differential mortality is taken into account. Modernizing the old-age tax means modernizing its base to match the peers it is measured against — while protecting benefits.

Sources

`SRC-SSA-TRUSTEES-2026`, `SRC-SSA-OACT`, `SRC-SSA-SOLVENCY`, `SRC-OECD-PENSIONS-2025`, `SRC-PENSION-CONTRIB-RATES`, `SRC-CHETTY-2016`, `SRC-NAS-2015`, `SRC-OMB-HIST-3-2-FY2027`, `SRC-OMB-HIST-2-4-FY2027`. All in `docs/sources/source-version-ledger.md`.